

G-CAF

GovDigital Composable Architecture Framework

Volume 19

Catalog Governance & Reporting Extensions

v2.0

Table of Contents

PART A — CATALOG EXTENSION (v1.18+)

Chapter 1 Cataloging Framework Overview

Meta-level governance for all G-CAF catalogs

Chapter 2 Catalog Metamodel

Standard structure, fields, and behaviors for every catalog

Chapter 3 Catalog Governance Model

ARB→DDA→Catalog Owner hierarchy and RACI

Chapter 4 Quality Rules (Gold/Silver/Bronze)

3-dimension quality scoring: completeness, accuracy, freshness

Chapter 5 Entry Lifecycle

State machine: Draft→Pending→Active→Deprecated→Retired

Chapter 6 Cross-Catalog Linking

Traceability across PBC→API→Data→Technology

Chapter 7 IDP Integration Patterns

Automation-first catalog maintenance via CI/CD pipelines

Chapter 8 3 Catalog Contracts

Provider, Consumer, and Quality contracts

PART B — REPORTING EXTENSION

Chapter 9 Reporting Framework

Report governance layers, adoption dashboard, framework health

Chapter 10 Report Contracts & Governance

Quality, security, and compliance requirements for reports

PART C — IDP EXTENSION

Chapter 11 Golden Path & Scorecard Specifications

Template standards and automated health assessments

Chapter 1

Cataloging Framework Overview

While individual catalogs (PBC, Technology, Data, Integration) define what they contain, this framework defines how they are structured, maintained, quality-governed, and linked.

Capability	Purpose
Unified Metamodel	Standard structure for all catalogs — common fields, behaviors, schemas
Quality Governance	Gold/Silver/Bronze quality tiers with scoring and SLAs
Lifecycle Management	Unified entry state machine across all catalogs
Cross-Catalog Linking	Explicit relationships: PBC → owns → APIs, uses → Technologies
Automation Standards	IDP/CI/CD integration for automated catalog maintenance

Chapter 2

Catalog Metamodel

Every G-CAF catalog implements this metamodel — the model of models.

Catalog Entity

Field	Type	Description
catalogId	String	Unique catalog identifier
name	String	Human-readable catalog name
owner	String	Responsible team/individual
qualityTier	Gold Silver Bronze	Quality governance level
schema	CatalogSchema	Entry field definitions
entries	CatalogEntry[]	Catalog content

CatalogEntry Entity

Field	Type	Description
entryId	UUID	System-generated unique ID
canonicalId	String	Human-readable ID (e.g., PBC-NOTIFY-001)
name	String	Entry display name
status	Enum	Draft, Pending, Active, Deprecated, Retired
owner	Reference	Entry steward
qualityScore	0–100	Computed quality rating
links	CatalogLink[]	Cross-catalog relationships

Chapter 3

Catalog Governance Model

Level	Body	Responsibilities
Strategic	ARB	Metamodel changes, new catalog approval, cross-catalog integration standards
Content	DDA	Entry approval/rejection, deprecation decisions, quality tier assignments
Operational	Catalog Owner	Day-to-day maintenance, entry steward assignment, quality monitoring
Entry	Entry Steward	Individual entry accuracy, freshness, completeness

Chapter 4

Quality Rules (Gold/Silver/Bronze)

Quality Score = (Completeness x 0.4) + (Accuracy x 0.3) + (Freshness x 0.3). Each dimension scored 0–100.

Tier	Score Range	Completeness SLA	Accuracy SLA	Freshness SLA
Gold	90–100	All fields populated; descriptions fully complete	Quarterly review; zero known errors	Updated within 30 days of any change
Silver	70–89	Required fields complete; optional fields populated	Semi-annual review; errors flagged	Updated within 90 days
Bronze	50–69	Required fields only	Annual review	Updated within 180 days
Below Bronze	<50	Incomplete required fields	Unreviewed or known errors	Stale (>180 days)

Below Bronze triggers remediation: entry flagged, steward notified, 30-day deadline to improve or entry deprecated.

Chapter 5

Entry Lifecycle

State	Description	Allowed Transitions
Draft	Entry being prepared; not visible to consumers	Submit → Pending Approval
Pending Approval	Awaiting DDA review	Approve → Active; Reject → Rejected
Active	Live entry; visible and usable	Deprecate → Deprecated
Deprecated	Marked for retirement; consumers notified	Retire → Retired (when zero consumers)
Retired	Archived; read-only historical record	None (terminal state)
Rejected	Failed review; revision required	Revise → Draft

Chapter 6

Cross-Catalog Linking

Links answer: Which APIs does this PBC expose? Which data entities does it own? Which PBCs use this technology? What is the impact of deprecating this API?

Link Type	Source Catalog	Target Catalog	Example
owns	PBC Catalog	Data Catalog	PBC-NOTIFY-001 owns DATA-MSG-001
exposes	PBC Catalog	Integration Catalog	PBC-NOTIFY-001 exposes API-NOTIFY-v1
uses	PBC Catalog	Technology Catalog	PBC-NOTIFY-001 uses TECH-NODEJS-18
consumes	PBC Catalog	PBC Catalog	PBC-CASE-001 consumes PBC-NOTIFY-001
implements	PBC Catalog	RA Patterns	PBC-FRAUD-001 implements PAT-CQRS

Links are validated: both source and target must exist, cardinality constraints enforced, orphan detection runs automatically. Impact analysis: before deprecating any entry, the system traces all incoming links to identify affected consumers.

Chapter 7

IDP Integration Patterns

Automation-first: manual catalog entry is the exception, not the rule.

Pattern	Trigger	Automation
CI/CD Pipeline Sync	Code merge to main	Auto-update catalog entry from catalog-info.yaml in repo
Gateway Event	Gateway decision	Auto-transition entry status (e.g., Go-Live → Active)
Schema Registry Sync	API/Event schema published	Auto-create/update Integration Catalog entry
Monitoring Feed	Health check data	Auto-update operational metrics in entry
Dependency Scan	Build pipeline	Auto-update cross-catalog links from detected dependencies

Implementation: 4-phase roadmap — Foundation (weeks 1–4: deploy platform, configure auth, set up event bus), Migration (weeks 5–8: extract from legacy, transform, load with validation), Integration (weeks 7–8: IDP pipelines, event subscriptions), Optimization (ongoing: quality monitoring, link validation, automation expansion).

Chapter 8

3 Catalog Contracts

Contract	Parties	Key Obligations
Provider Contract	Catalog Owners, Entry Stewards, Data Owners	CRUD operations via Entry API; maintain quality scores; respond to consumer queries
Consumer Contract	Developers, Architects, System Integrators	Searching catalog via Search API (REST/GraphQL); retrieve details; export bulk data
Quality Contract	Both Provider and Consumer	Quality Score API for metrics; quality rules management; validation checks; feedback loops

Chapter 9

Reporting Framework

Standardized reporting integrated with G-CAF governance — measuring adoption, compliance, and framework health.

G-CAF Adoption Dashboard

Attribute	Value
Report ID	REP-GCAF-ADOPTION-001
Frequency	Monthly
Audience	Executive, ARB, Governance
Content	Principle compliance, gateway reviews, RA adoption, PBC catalog maturity
Approver	ARB

Framework Health Report

Metric	Value	Status
Total Modules	473	Healthy
Frontmatter Coverage	99.8%	Healthy
Module ID Coverage	99.8%	Healthy
Audience Coverage	21.1%	Needs improvement
Graph Coverage	81.6%	Adequate
Principles	19	Complete
Contracts	8	Complete
Reference Architectures	10	Growing

Report Contracts & Governance

4-Layer Report Governance

Layer	Requirement
Layer 1: Principles	Privacy by Design (no PII without consent), Security by Design (classification enforced), Observable by Default
Layer 2: Contracts	Report Data Contract (handling, retention, access), Report Quality Contract (accuracy, timeliness, completeness)
Layer 3: Gateways	New/changed reports pass Design Gateway (design review) and Go-Live Gateway (production readiness)
Layer 4: Approval	Executive reports → ARB; Entity reports → DDA; Team reports → automated + DDA notification

Report contracts include: Report Registry (REST), Data Lineage (GraphQL), Quality Check (REST), and Access Control (OAuth 2.0). All reports must have documented data lineage before publication.

Chapter 11

Golden Path & Scorecard Specifications

Golden Paths

Principle	Description
Opinionated	Make decisions for developers; don't expose every option
Paved	Clear, well-maintained path with guardrails
Complete	Everything needed to reach production
Compliant	G-CAF requirements built-in from scaffolding
Escapable	Developers can customize after initial scaffolding

Without golden paths: 2 weeks setup, inconsistent results. With golden paths: 2 days, standardized compliance.

Scorecards

Scorecard	Focus	Key Metrics
Security Scorecard	Security posture	Vulnerability count, scan freshness, secret exposure, RBAC compliance
Reliability Scorecard	Operational health	Uptime, error rate, p99 latency, health check status
Documentation Scorecard	Documentation quality	API docs coverage, runbook exists, architecture diagrams current
Dependencies Scorecard	Supply chain health	Outdated dependencies, vulnerability count, SBOM freshness
Operations Scorecard	Operational readiness	Monitoring configured, alerting active, backup verified, DR tested
Value Realization Scorecard	Business value	Consumer count, reuse rate, gateway compliance, SLA achievement

Scorecards provide automated, objective PBC health assessments. Replace subjective evaluations with data-driven metrics. Enable continuous improvement and governance at scale. Scores visible in IDP portal for all stakeholders.